

Analytical and Numerical Solutions to the Black-Scholes Equation

Conceptual Foundations, Exact Derivations, and Monte Carlo Methods

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Abstract

This report presents a comprehensive theoretical and computational framework for the valuation of European options under the Black-Scholes model. Recognizing the non-differentiable, stochastic nature of asset trajectories, we utilize Geometric Brownian Motion and Ito's Lemma to construct a continuously hedged, risk-neutral portfolio, culminating in the derivation of the Black-Scholes Partial Differential Equation (PDE). To obtain an exact closed-form solution, the PDE is algebraically mapped to the fundamental 1D Heat Equation via a sequence of domain transformations. By applying spatial Fourier Transforms to the Cauchy problem and leveraging the sifting property of the Dirac delta distribution, we mathematically prove that the initial boundary condition resolves as a spatial convolution against the Gaussian Heat Kernel. Finally, this theoretical architecture is empirically validated through a vectorized Monte Carlo simulation in Python. By integrating the risk-neutral random walk, the computational model demonstrates aggressive asymptotic convergence to the exact analytical price in strict accordance with the Law of Large Numbers.

1 Introduction and Mathematical Foundations

The objective of this project is to explore the theoretical pricing of financial derivatives and implement a computational framework to approximate their fair value. Classical deterministic mathematics is fundamentally ill-equipped to model financial markets due to the inherently stochastic, unpredictable nature of asset prices. Therefore, this report constructs a rigorous continuous-time pricing model from the ground up.

- **Section 1** establishes the necessary stochastic calculus, specifically Geometric Brownian Motion and Ito's Lemma, to model jagged market dynamics.
- **Section 2** leverages these mathematical tools to construct a risk-neutral, self-financing portfolio, applying the economic principle of no-arbitrage to derive the foundational Black-Scholes Partial Differential Equation (PDE).
- **Section 3** demonstrates the exact closed-form analytical solution by transforming the financial PDE into the fundamental Heat Equation from physics.
- **Section 4** moves beyond theoretical proofs, implementing a Monte Carlo simulation in Python to solve the pricing model computationally via the Law of Large Numbers.

1.1 The Conceptual Problem with Market Dynamics

In classical mechanics or standard statistics, continuous variables are generally assumed to be smooth. If a particle is moving, its trajectory over a sufficiently small time interval, dt , looks like a straight line. Because of this smoothness, we can calculate a definitive derivative, or velocity.

Financial markets do not behave this way. If we observe the price of an asset, S_t , and continuously zoom in on smaller and smaller time frames, the path never becomes smooth; it remains infinitely jagged. This constant "jitter" is driven by the continuous flow of new information, trades, and market shocks.

Because the path is nowhere differentiable, we cannot simply write $\frac{dS}{dt}$. We must separate the asset's movement into a predictable trend and an unpredictable shock.

1.2 Constructing the Asset Engine: Geometric Brownian Motion

To model this dynamically, we construct the asset price movement conceptually step-by-step. A naive approach might be to model the absolute dollar change of the asset, dS_t , directly. However, if we modeled $dS_t = \mu dt + \sigma dW_t$ (Standard Brownian Motion), a severe negative random shock could easily drive the stock price below zero, which violates the reality of limited liability in equity markets.

To fix this, we must instead model the *instantaneous percentage return* of the asset over an infinitesimally small time step dt , defined mathematically as $\frac{dS_t}{S_t}$.

This percentage return is conceptually driven by two distinct forces:

1. **The Deterministic Drift:** In the absence of any random news, the asset should grow at a risk-adjusted rate. We call this constant rate μ . Over time dt , this contributes a guaranteed expected return of μdt .
2. **The Stochastic Diffusion (The Foundational Assumption):** To account for the "jitter," we make a strict mathematical assumption: we assume the market's random shocks behave perfectly like a standard Wiener process (or standard Brownian motion), denoted as W_t . By assuming this, we are asserting that market returns have no memory (independent increments) and that the shocks follow a perfect Normal distribution with a mean of zero and a variance equal to the time step. The increment of this assumed process, dW_t , represents our market shock. We scale the intensity of this shock by a constant volatility parameter, σ , contributing a random return of σdW_t .

Combining these conceptual blocks, we arrive at the Stochastic Differential Equation (SDE) known as Geometric Brownian Motion (GBM):

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t \implies dS_t = \mu S_t dt + \sigma S_t dW_t \quad (1)$$

By formatting the equation to define the percentage return ($\frac{dS_t}{S_t}$), we ensure that the absolute dollar shock ($\sigma S_t dW_t$) is strictly proportional to the current price S_t . If the stock price approaches zero, the absolute magnitude of the random shock shrinks proportionally. This guarantees that the asset price approaches zero asymptotically but can never cross it to become negative.

1.3 The Step-by-Step Mechanics of Ito's Lemma

Now we must price a derivative contract, such as an option, whose value V depends on the current asset price and time: $V(S, t)$. Because S is a stochastic variable, we need a way to track

how V changes as S and t change.

In standard calculus, we would use a Taylor series expansion to find the differential dV . For a small change in S (dS) and t (dt), the expansion begins as:

$$dV = \frac{\partial V}{\partial t}dt + \frac{\partial V}{\partial S}dS + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}(dS)^2 + \dots \quad (2)$$

In standard mathematics, as $dt \rightarrow 0$, the $(dS)^2$ term becomes infinitesimally small compared to dS and is therefore discarded. However, this is the exact moment where standard calculus fails in finance.

We must conceptually evaluate the quadratic variation of our stochastic process, $(dS)^2$. Let us substitute our GBM equation into this squared term:

$$(dS)^2 = (\mu S dt + \sigma S dW_t)^2 = \mu^2 S^2 (dt)^2 + 2\mu\sigma S^2 (dt)(dW_t) + \sigma^2 S^2 (dW_t)^2 \quad (3)$$

As we take the limit $dt \rightarrow 0$:

- $(dt)^2$ shrinks to zero much faster than dt , so it vanishes.
- $(dt)(dW_t)$ also shrinks to zero and vanishes.
- **The Critical Difference (The Normal Distribution):** The squared random shock, $(dW_t)^2$, does *not* vanish. By the formal definition of a Wiener process, the stochastic increment dW_t follows a Normal distribution with a mean of zero and a variance exactly equal to the time step dt . Mathematically, $dW_t \sim \mathcal{N}(0, dt)$. Because its variance is dt , the expected value of the squared shock is exactly dt ($\mathbb{E}[(dW_t)^2] = dt$). In the continuous-time limit, this expected value becomes a deterministic certainty, meaning the randomness cancels out and $(dW_t)^2 = dt$.

Therefore, the only term that survives the squaring process is the pure volatility component:

$$(dS)^2 = \sigma^2 S^2 dt \quad (4)$$

When we plug this conceptual realization back into our original Taylor series expansion, the second-derivative term survives. This gives us the foundational equation of stochastic calculus, Ito's Lemma:

$$dV = \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S \frac{\partial V}{\partial S} dW_t \quad (5)$$

Conceptually, this mathematically proves that the volatility of the underlying asset directly impacts the deterministic time-decay of the option's value. The dW_t term at the end perfectly mirrors the random shock of the underlying asset, which will allow us to construct a hedging strategy in the next section to entirely eliminate market risk.

2 Derivation of the Black-Scholes PDE

In this section, we utilize the mathematical tools developed in Section 1—specifically Geometric Brownian Motion and Ito's Lemma—to construct a continuous-time hedging strategy. By enforcing the principle of no-arbitrage, we eliminate the stochastic component of the asset's price movement and derive the Black-Scholes Partial Differential Equation.

2.1 Constructing the Hedged Portfolio

The fundamental insight of the Black-Scholes model is that it is possible to continuously construct a locally risk-free portfolio consisting of the derivative and the underlying asset.

Let us define a self-financing portfolio, Π , constructed at time t by holding exactly one unit of the derivative contract $V(S, t)$ and shorting Δ units of the underlying asset S . The value of this portfolio is:

$$\Pi = V - \Delta S \quad (6)$$

Over an infinitesimally small time step dt , the change in the portfolio's value, $d\Pi$, is given by the change in the option's value minus the change in the underlying asset's value:

$$d\Pi = dV - \Delta dS \quad (7)$$

2.2 Applying Ito's Lemma and Eliminating Risk

From Section 1, we know the dynamics of the underlying asset follow Geometric Brownian Motion ($dS = \mu S dt + \sigma S dW_t$), and the change in the option's value is given by Ito's Lemma. We substitute these into our portfolio equation:

$$d\Pi = \left[\left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S \frac{\partial V}{\partial S} dW_t \right] - \Delta (\mu S dt + \sigma S dW_t)$$

We now gather the deterministic (dt) and stochastic (dW_t) terms to isolate the sources of return and risk:

$$d\Pi = \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - \mu S \Delta \right) dt + \sigma S \left(\frac{\partial V}{\partial S} - \Delta \right) dW_t \quad (8)$$

The term multiplied by dW_t represents the market risk. To construct a truly risk-free portfolio, we must eliminate this stochastic randomness. We achieve this by continuously choosing our hedge ratio, Δ , such that the dW_t coefficient equals zero:

$$\Delta = \frac{\partial V}{\partial S} \quad (9)$$

By setting Δ equal to the option's sensitivity to the underlying asset (the "Delta" of the option), the dW_t term perfectly cancels out. The change in the portfolio value becomes entirely deterministic:

$$d\Pi = \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt \quad (10)$$

Notice that the physical drift rate, μ , has also been mathematically eliminated from the equation. This implies that the fair value of an option is entirely independent of the investor's expectations regarding the stock's future directional growth.

2.3 The No-Arbitrage Argument (The Bank Account Analogy)

Since the portfolio Π is now entirely free of market risk, the economic principle of no-arbitrage dictates that it must yield exactly the same return as any other risk-free investment. If this portfolio grew faster or slower than a standard risk-free asset, high-frequency traders could exploit the difference to generate infinite, guaranteed profit (arbitrage).

To mathematically prevent this arbitrage, the market forces the portfolio to grow at the exact same continuous rate as money sitting in a bank account earning a risk-free interest rate, r . Just as a bank balance grows by the equation $d(\text{Balance}) = r \times (\text{Balance}) \times dt$, the deterministic growth of our hedged portfolio must satisfy:

$$d\Pi = r\Pi dt = r \left(V - \frac{\partial V}{\partial S} S \right) dt \quad (11)$$

2.4 The Final Black-Scholes PDE

We now have two equations describing the identical risk-free change in the portfolio, $d\Pi$. We equate Equation (10) and Equation (11):

$$\left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt = r \left(V - \frac{\partial V}{\partial S} S \right) dt \quad (12)$$

Dividing both sides by the time differential dt and rearranging the terms to isolate zero on one side yields the final, continuous-time Black-Scholes Partial Differential Equation:

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0 \quad (13)$$

This second-order linear parabolic partial differential equation must hold true for any derivative whose value depends solely on the underlying asset S and time t . The specific type of derivative (e.g., Call vs. Put) is determined entirely by the boundary and terminal conditions applied when solving the PDE.

3 Analytical Solution: Exact Mathematical Derivation

The Black-Scholes PDE (Equation 13) is a linear parabolic equation with variable coefficients (S and S^2). To solve this differential equation analytically, we must apply a sequence of domain transformations to eliminate the variable coefficients, followed by an integrating factor substitution to reduce the equation to the fundamental 1D Heat Equation.

3.1 Step 1: Domain Transformation (Chain Rule Expansion)

We introduce two new independent variables. Let x represent the logarithmic moneyness of the asset, and let τ represent the time to maturity scaled by variance:

$$x = \ln \left(\frac{S}{K} \right) \implies S = K e^x \quad (14)$$

$$\tau = \frac{\sigma^2}{2} (T - t) \implies t = T - \frac{2\tau}{\sigma^2} \quad (15)$$

We define a new function $v(x, \tau)$ such that $V(S, t) = K \cdot v(x, \tau)$. To substitute this into the PDE, we must compute the partial derivatives of V with respect to t and S using the chain rule.

Time Derivative:

$$\frac{\partial V}{\partial t} = K \frac{\partial v}{\partial \tau} \frac{\partial \tau}{\partial t} = K \frac{\partial v}{\partial \tau} \left(-\frac{\sigma^2}{2} \right) = -K \frac{\sigma^2}{2} \frac{\partial v}{\partial \tau} \quad (16)$$

First Spatial Derivative:

$$\frac{\partial V}{\partial S} = K \frac{\partial v}{\partial x} \frac{\partial x}{\partial S} = K \frac{\partial v}{\partial x} \left(\frac{1}{S} \right) \quad (17)$$

Second Spatial Derivative (using the Product Rule):

$$\frac{\partial^2 V}{\partial S^2} = K \frac{\partial}{\partial S} \left(\frac{1}{S} \frac{\partial v}{\partial x} \right) = K \left(-\frac{1}{S^2} \frac{\partial v}{\partial x} + \frac{1}{S} \frac{\partial^2 v}{\partial x^2} \frac{\partial x}{\partial S} \right) = \frac{K}{S^2} \left(\frac{\partial^2 v}{\partial x^2} - \frac{\partial v}{\partial x} \right) \quad (18)$$

We now substitute these explicitly derived partials back into the original Black-Scholes PDE:

$$\left[-K \frac{\sigma^2}{2} \frac{\partial v}{\partial \tau} \right] + \frac{1}{2} \sigma^2 S^2 \left[\frac{K}{S^2} \left(\frac{\partial^2 v}{\partial x^2} - \frac{\partial v}{\partial x} \right) \right] + rS \left[\frac{K}{S} \frac{\partial v}{\partial x} \right] - r(Kv) = 0 \quad (19)$$

Notice how the variable coefficients perfectly cancel out ($S^2 \cdot \frac{1}{S^2} = 1$ and $S \cdot \frac{1}{S} = 1$). Dividing the entire equation by the constant $K \frac{\sigma^2}{2}$ yields:

$$-\frac{\partial v}{\partial \tau} + \left(\frac{\partial^2 v}{\partial x^2} - \frac{\partial v}{\partial x} \right) + \frac{2r}{\sigma^2} \frac{\partial v}{\partial x} - \frac{2r}{\sigma^2} v = 0 \quad (20)$$

Letting $k = \frac{2r}{\sigma^2}$ and isolating the time derivative, we arrive at a constant-coefficient convection-diffusion equation:

$$\frac{\partial v}{\partial \tau} = \frac{\partial^2 v}{\partial x^2} + (k - 1) \frac{\partial v}{\partial x} - kv \quad (21)$$

Conceptual Remark: The original equation is mathematically difficult because the parameters constantly shift depending on the stock price (S). By redefining our variable using the natural logarithm ($x = \ln(S/K)$), we map the unpredictable stock market onto a flat, infinite number line where the strike price sits perfectly at zero. By applying this translation via the Chain Rule, the messy S and S^2 terms completely vanish.

3.2 Step 2: Eliminating the Advection and Decay Terms

Equation (21) contains a first-derivative advection term ($(k - 1)v_x$) and a zeroth-derivative decay term ($-kv$). We eliminate these by introducing an integrating factor. We define a new function $u(x, \tau)$ with an exponential wrapper containing undetermined constants α and β :

$$v(x, \tau) = e^{\alpha x + \beta \tau} u(x, \tau) \quad (22)$$

We compute the partial derivatives of v utilizing the product rule:

$$v_\tau = e^{\alpha x + \beta \tau} (\beta u + u_\tau) \quad (23)$$

$$v_x = e^{\alpha x + \beta \tau} (\alpha u + u_x) \quad (24)$$

$$v_{xx} = e^{\alpha x + \beta \tau} (\alpha^2 u + 2\alpha u_x + u_{xx}) \quad (25)$$

Substituting these into Equation (21) and dividing out the common $e^{\alpha x + \beta \tau}$ term:

$$\beta u + u_\tau = (\alpha^2 u + 2\alpha u_x + u_{xx}) + (k - 1)(\alpha u + u_x) - kv \quad (26)$$

Grouping the terms by the derivatives of u :

$$u_\tau = u_{xx} + u_x[2\alpha + (k - 1)] + u[\alpha^2 + \alpha(k - 1) - k - \beta] \quad (27)$$

To reduce this to the Heat Equation ($u_\tau = u_{xx}$), the coefficients of u_x and u must strictly equal zero. This provides a system of equations to solve for α and β :

$$2\alpha + k - 1 = 0 \implies \alpha = -\frac{1}{2}(k - 1) \quad (28)$$

$$\beta = \alpha^2 + \alpha(k-1) - k \quad (29)$$

$$= \left[-\frac{1}{2}(k-1) \right]^2 + \left[-\frac{1}{2}(k-1) \right] (k-1) - k \quad (30)$$

$$= -\frac{1}{4}(k-1)^2 - k \quad (31)$$

$$= -\frac{1}{4}(k^2 - 2k + 1 + 4k) \quad (32)$$

$$= -\frac{1}{4}(k+1)^2 \quad (33)$$

By selecting these specific constants, the complex financial parameters are completely neutralized, and the PDE algebraically collapses into the 1D Heat Equation:

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2} \quad (34)$$

Conceptual Remark: Equation (21) still contains the natural upward "drift" of the market and the "decay" of money due to the interest rate. By wrapping our equation in an exponential function ($e^{\alpha x + \beta \tau}$), we create a mathematical "treadmill" running backward at the exact same speed as the market's natural drift. Once this treadmill neutralizes the financial variables, the Black-Scholes formula elegantly collapses into the 1D Heat Equation.

3.3 Step 3: The Fourier Transform and The Cauchy Problem

Having cleanly reduced the Black-Scholes PDE to the 1D Heat Equation ($u_\tau = u_{xx}$), we must discover its fundamental solution. We seek the mathematical behavior of a single, infinitely concentrated point of certainty diffusing over time. In mathematics, this is expressed as a Cauchy problem with a Dirac delta function ($\delta(x)$) as the initial condition:

$$\begin{cases} u_\tau = u_{xx} & \text{for } x \in \mathbb{R}, \tau > 0 \\ u(x, 0) = \delta(x) & \text{for } x \in \mathbb{R} \end{cases} \quad (35)$$

Conceptual Remark: Solving a second-order partial differential equation directly in the spatial domain is highly complex. To bypass this, we apply a spatial Fourier Transform to shift the entire equation into the frequency domain.

Let $\hat{u}(\omega, \tau)$ be the Fourier transform of $u(x, \tau)$. We apply the mathematical definition of the transform:

$$\hat{u}(\omega, \tau) = \mathcal{F}\{u(x, \tau)\} = \int_{-\infty}^{\infty} u(x, \tau) e^{-i\omega x} dx \quad (36)$$

Conceptual Remark: The term $e^{-i\omega x}$ acts as a mathematical prism. By integrating our function against this complex exponential, we strip away the spatial variable x and replace it with the frequency variable ω .

Applying this transformation to both sides of the Heat Equation ($u_\tau = u_{xx}$):

$$\frac{\partial \hat{u}}{\partial \tau} = (i\omega)^2 \hat{u} = -\omega^2 \hat{u} \quad (37)$$

Conceptual Remark: This is the crux of the derivation. A known property of the Fourier Transform is that taking a spatial double derivative ($\frac{\partial^2}{\partial x^2}$) in the real world is mathematically identical to simply multiplying by $(i\omega)^2$ in the frequency domain. Because $i^2 = -1$, the hard calculus vanishes, leaving only basic algebra ($-\omega^2$).

This operation brilliantly collapses the difficult PDE into a basic first-order Ordinary Differential Equation (ODE). The solution to this ODE is straightforward:

$$\hat{u}(\omega, \tau) = A(\omega)e^{-\omega^2\tau} \quad (38)$$

To find the arbitrary constant $A(\omega)$, we evaluate our initial condition. The Fourier transform of the Dirac delta function is simply 1:

$$\hat{u}(\omega, 0) = \mathcal{F}\{\delta(x)\} = 1 \implies A(\omega) = 1 \quad (39)$$

Conceptual Remark: Because the Fourier transform of our single-point starting condition $\delta(x)$ evaluates cleanly to 1, our final frequency-domain solution is a simple exponential decay: $\hat{u} = e^{-\omega^2\tau}$.

To find our final spatial solution, we apply the **Inverse Fourier Transform**:

$$u(x, \tau) = \mathcal{F}^{-1}\{e^{-\omega^2\tau}\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\omega^2\tau} e^{i\omega x} d\omega \quad (40)$$

Evaluating the Inverse Transform:

To solve this integral, we consolidate the exponent and complete the square with respect to the frequency variable ω :

$$-\tau\omega^2 + ix\omega = -\tau \left(\omega^2 - \frac{ix}{\tau}\omega \right) = -\tau \left(\omega - \frac{ix}{2\tau} \right)^2 - \frac{x^2}{4\tau} \quad (41)$$

Substituting this completed square back into the integral, the real-valued term independent of ω factors completely outside the integral operation:

$$u(x, \tau) = e^{-\frac{x^2}{4\tau}} \cdot \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\tau(\omega - \frac{ix}{2\tau})^2} d\omega \quad (42)$$

By applying the contour-shifted substitution $z = \sqrt{\tau}(\omega - \frac{ix}{2\tau})$, where $d\omega = \frac{dz}{\sqrt{\tau}}$, the equation collapses into the standard Gaussian integral ($\int_{-\infty}^{\infty} e^{-z^2} dz = \sqrt{\pi}$):

$$u(x, \tau) = e^{-\frac{x^2}{4\tau}} \left(\frac{1}{2\pi} \cdot \frac{\sqrt{\pi}}{\sqrt{\tau}} \right) = \frac{1}{\sqrt{4\pi\tau}} e^{-\frac{x^2}{4\tau}} \equiv \Phi(\mathbf{x}, \tau) \quad (43)$$

Conceptual Remark: By applying the Inverse Fourier Transform to return to the real world, the equation resolves through a standard Gaussian integration. This proves rigorously that a deterministic point source diffuses exactly as a Normal Probability Density Function over time.

Rigorous Derivation of the Convolution Integral

Before executing the proof for the continuous option payoff, we must rigorously define the mathematical components of our initial condition framework. In financial terms, we are transitioning from a single theoretical point source to the entire spectrum of future asset prices:

- **The Dirac Delta Distribution $\delta(x - s)$:** A generalized mathematical function acting as a strictly localized point source or "filter." It is exactly zero everywhere except when the scanning variable s perfectly equals the target variable x , where it spikes to infinity.
- **The State Space Function $f(s)$:** The continuous mathematical function representing the entire Call option payoff evaluated across all possible future asset prices (s).

- **The Target Value $f(x)$:** The exact, singular, deterministic value of the option payoff evaluated exclusively at our specific target coordinate (x) .

To mathematically prove why the final solution manifests strictly as a convolution, we use these definitions to invoke the **sifting property** of the Dirac delta distribution. Any continuous function $f(x)$ can be exactly represented as an infinite integral of its own shifted, isolated point sources:

$$f(x) = \int_{-\infty}^{\infty} f(s)\delta(x-s)ds \quad (44)$$

Proof of the Sifting Property:

To prove this equality, we rely on the formal definition of the Dirac delta distribution, which is established by two core properties: 1. $\delta(x-s) = 0$ for all $s \neq x$. 2. $\int_{-\infty}^{\infty} \delta(x-s)ds = 1$.

Because $\delta(x-s)$ is exactly zero everywhere except at the singular coordinate $s = x$, multiplying it by our continuous payoff $f(s)$ inside the integral ensures that the integrand evaluates to zero everywhere except at exactly x . Therefore, inside the bounds of the integral, $f(s)$ effectively becomes a constant evaluated at $f(x)$. Since it behaves as a constant, we can factor $f(x)$ strictly outside of the integration with respect to s :

$$\int_{-\infty}^{\infty} f(s)\delta(x-s)ds = f(x) \int_{-\infty}^{\infty} \delta(x-s)ds \quad (45)$$

By the second defining property of the Dirac delta, the remaining integral strictly evaluates to 1. The equation therefore cleanly resolves to $f(x) \cdot 1 = f(x)$. This formally proves that a continuous shape is mathematically equivalent to the infinite integral sum of its discrete, isolated point sources.

Operator Notation and the Boundary Value Problem

To rigorously define the inverse operator $\mathcal{L}^{-1}$, we must clarify the nature of the Cauchy problem. The notation $\mathcal{L}[u] = 0$ represents the homogeneous evolution of the system for $\tau > 0$, dictating that no external probability or heat is injected into the system after the initial state. It defines *how* the system evolves, not *where* it starts.

The initial condition $f(x)$ exists strictly at $\tau = 0$. Therefore, when we write $u(x, \tau) = \mathcal{L}^{-1}[f(x)]$, we are not performing algebraic division. We are using standard operator shorthand to denote the application of the Green's Function. $\mathcal{L}^{-1}$ represents the integral operator that maps the initial boundary condition forward in time via convolution with the fundamental solution.

Applying the Solution Operator (Linearity)

By definition, the final solution $u(x, \tau)$ to the Cauchy problem is found by applying the inverse solution operator, $\mathcal{L}^{-1}$, to the initial boundary condition $f(x)$. Mathematically, this is expressed as $u(x, \tau) = \mathcal{L}^{-1}[f(x)]$.

By substituting our integral representation of $f(x)$ derived from the sifting property directly into this operator argument, we arrive at the foundational setup for our convolution:

$$u(x, \tau) = \mathcal{L}^{-1} \left[\int_{-\infty}^{\infty} f(s)\delta(x-s)ds \right] \quad (46)$$

Let $\mathcal{L} = \frac{\partial}{\partial \tau} - \frac{\partial^2}{\partial x^2}$ define the linear differential operator of the Heat Equation. The fundamental solution $\Phi(x, \tau)$ satisfies $\mathcal{L}[\Phi] = 0$ with the initial condition $\Phi(x, 0) = \delta(x)$. Because the

domain is spatially invariant, shifting the initial condition to $\delta(x - s)$ exactly shifts the solution to $\Phi(x - s, \tau)$.

Because the differential operator $\mathcal{L}$ is strictly linear, it obeys the principle of superposition. The solution to an integral of initial conditions is equal to the integral of their respective solutions. This mathematical linearity allows us to pass the solution operator directly inside the integral. Since $f(s)$ acts as a constant with respect to the operator, $\mathcal{L}^{-1}$ attaches exclusively to the Dirac delta:

$$u(x, \tau) = \int_{-\infty}^{\infty} f(s) \mathcal{L}^{-1}[\delta(x - s)] ds = \int_{-\infty}^{\infty} f(s) \Phi(x - s, \tau) ds \quad (47)$$

This formally proves that the solution to the Cauchy problem is the spatial convolution of the initial condition with the fundamental solution ($u = f * \Phi$).

Evaluating the Option Payoff

We define our true initial condition, $f(x)$, by taking the known European Call option payoff at expiration ($V(S, T) = \max(S - K, 0)$) and applying our variable substitutions from Steps 1 and 2:

$$S = Ke^x \quad \text{and} \quad V(S, t) = K \cdot e^{\alpha x + \beta \tau} u(x, \tau) \quad (48)$$

At the exact moment of expiration ($\tau = 0$), the β term vanishes because $e^{\beta(0)} = 1$:

$$V(S, T) = K \cdot e^{\alpha x}(1) \cdot u(x, 0) \implies Ke^{\alpha x} u(x, 0) = K \max(e^x - 1, 0) \quad (49)$$

Dividing both sides by $Ke^{\alpha x}$, we isolate our true starting shape, $f(x)$:

$$\mathbf{f}(\mathbf{x}) = \mathbf{e}^{-\alpha \mathbf{x}} \max(\mathbf{e}^{\mathbf{x}} - 1, \mathbf{0}) \quad (50)$$

Substituting our explicit $f(s)$ payoff into the derived convolution integral, we truncate the lower bound to 0 because the Call option is worthless (zero) on the negative interval:

$$u(x, \tau) = \frac{1}{\sqrt{4\pi\tau}} \int_0^{\infty} e^{-\alpha s} (e^s - 1) e^{-\frac{(x-s)^2}{4\tau}} ds \quad (51)$$

3.4 Step 4: Splitting the Integral and Explicit Algebraic Expansion

To evaluate this convolution, we expand the $(e^s - 1)$ bracket. This mathematically splits the Call option into its two components: the probability of receiving the stock (I_1) minus the probability of paying the strike (I_2).

$$u(x, \tau) = I_1 - I_2 \quad (52)$$

$$I_1 = \frac{1}{\sqrt{4\pi\tau}} \int_0^{\infty} e^{(1-\alpha)s - \frac{(s-x)^2}{4\tau}} ds \quad (53)$$

$$I_2 = \frac{1}{\sqrt{4\pi\tau}} \int_0^{\infty} e^{-\alpha s - \frac{(s-x)^2}{4\tau}} ds \quad (54)$$

We will solve I_2 explicitly to demonstrate the derivation of the d_2 variable. To solve Equation (54), we must consolidate the exponential terms into a single quadratic structure by **Completing the Square**.

First, we expand the numerator of the fraction inside the exponent:

$$-\alpha s - \frac{s^2 - 2sx + x^2}{4\tau} = -\frac{s^2 - 2s(x - 2\alpha\tau) + x^2}{4\tau} \quad (55)$$

We isolate the terms containing s to complete the square: $s^2 - 2s(x - 2\alpha\tau)$. To make this a perfect square, we add and subtract $(x - 2\alpha\tau)^2$:

$$[s - (x - 2\alpha\tau)]^2 - (x - 2\alpha\tau)^2 \quad (56)$$

Substituting this perfect square back into our full exponent:

$$-\frac{[s - (x - 2\alpha\tau)]^2}{4\tau} + \frac{(x - 2\alpha\tau)^2 - x^2}{4\tau} \quad (57)$$

The second fraction does not contain the variable s , meaning it acts as a constant with respect to our integral. We can expand and simplify this constant term:

$$\frac{x^2 - 4\alpha\tau x + 4\alpha^2\tau^2 - x^2}{4\tau} = \frac{-4\alpha\tau x + 4\alpha^2\tau^2}{4\tau} = -\alpha x + \alpha^2\tau \quad (58)$$

We pull this constant term outside of the integral, leaving only the perfect square inside:

$$I_2 = e^{-\alpha x + \alpha^2\tau} \cdot \frac{1}{\sqrt{4\pi\tau}} \int_0^\infty e^{-\frac{[s - (x - 2\alpha\tau)]^2}{4\tau}} ds \quad (59)$$

The z -Substitution (Deriving d_2): To evaluate the remaining integral, we execute a variable substitution to translate the exponent into the format of a Standard Normal Distribution ($-z^2/2$). We let:

$$z = \frac{s - (x - 2\alpha\tau)}{\sqrt{2\tau}} \implies dz = \frac{ds}{\sqrt{2\tau}} \quad (60)$$

This substitution requires us to change the limits of integration. The upper limit (∞) remains ∞ . However, when evaluating the lower limit where $s = 0$, z becomes:

$$z_{\text{lower}} = \frac{-(x - 2\alpha\tau)}{\sqrt{2\tau}} = -d_2 \quad (61)$$

Our integral now perfectly mirrors the Standard Normal Probability Density Function:

$$I_2 = e^{-\alpha x + \alpha^2\tau} \cdot \frac{1}{\sqrt{2\pi}} \int_{-d_2}^\infty e^{-z^2/2} dz \quad (62)$$

Because the normal distribution is perfectly symmetric, integrating from $-d_2$ to ∞ is mathematically equivalent to integrating from $-\infty$ to positive d_2 . This is the exact formal definition of the Cumulative Distribution Function, $\mathcal{N}(d_2)$. Therefore:

$$I_2 = e^{-\alpha x + \alpha^2\tau} \mathcal{N}(d_2) \quad (63)$$

A perfectly symmetrical derivation applied to I_1 yields $e^{(1-\alpha)x + (1-\alpha)^2\tau} \mathcal{N}(d_1)$. By substituting I_1 and I_2 back into our original framework and reversing the variable substitutions ($x \rightarrow S, \tau \rightarrow t$), the exponential wrappers cancel out entirely, leaving the final analytical formula for the European Call Option (C):

$$C(S, t) = S\mathcal{N}(d_1) - Ke^{-r(T-t)}\mathcal{N}(d_2) \quad (64)$$

4 Numerical Solutions: Monte Carlo Simulation

While the Black-Scholes Equation can be solved algebraically using the heat transformation outlined in Section 3, an alternative and highly robust computational approach relies on statistical simulation. Monte Carlo methods bypass the differential equation entirely, relying instead on the Law of Large Numbers to directly estimate the expected payoff of the derivative.

4.1 Integrating the Risk-Neutral Random Walk

The fundamental mathematical trick of the Black-Scholes framework is the concept of risk-neutral valuation. We demonstrated in Section 2 that the market risk (σdW_t) and the physical drift (μ) are eliminated when creating a hedged portfolio. Because the option's value is independent of the investor's risk preferences, we can pretend that we live in a risk-neutral world where all assets simply grow at the risk-free rate, r .

The risk-neutral stochastic differential equation for the asset is:

$$dS_t = rS_t dt + \sigma S_t dW_t \quad (65)$$

Due to the stochastic nature of dW_t , this equation cannot be integrated using standard calculus. We must apply Ito's Lemma to the natural logarithm of the stock price, $f(S, t) = \ln(S_t)$, yielding the partial derivatives $\frac{\partial f}{\partial S} = \frac{1}{S_t}$ and $\frac{\partial^2 f}{\partial S^2} = -\frac{1}{S_t^2}$.

Applying Ito's formula:

$$d(\ln S_t) = \left(\frac{1}{S_t}\right) dS_t - \frac{1}{2} \left(\frac{1}{S_t^2}\right) (dS_t)^2 \quad (66)$$

Substituting dS_t and the quadratic variation $(dS_t)^2 = \sigma^2 S_t^2 dt$:

$$d(\ln S_t) = (r dt + \sigma dW_t) - \frac{1}{2} \sigma^2 dt = \left(r - \frac{1}{2} \sigma^2\right) dt + \sigma dW_t \quad (67)$$

This equation expresses the change in the logarithmic return. The $-\frac{1}{2}\sigma^2$ term represents "volatility drag," demonstrating mathematically that variance degrades compound growth over time. Integrating both sides from time 0 to T yields:

$$\ln(S_T) - \ln(S_0) = \left(r - \frac{1}{2} \sigma^2\right) T + \sigma(W_T - W_0) \quad (68)$$

By the definition of a standard Wiener process, $W_0 = 0$ and $W_T \sim \mathcal{N}(0, T)$. We can express this as $\sqrt{T}Z$, where $Z \sim \mathcal{N}(0, 1)$ is a standard normal random variable. Exponentiating both sides provides the explicit exact solution for the stock price at maturity:

$$S_T = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2\right) T + \sigma \sqrt{T} Z \right) \quad (69)$$

4.2 The Law of Large Numbers

Because Z is a random variable, S_T is also a random variable with a log-normal distribution. To find the fair value of an option today, we execute the following statistical procedure:

1. **Simulate:** Use a computational random number generator to draw tens of thousands of different values for Z .

2. **Calculate Paths:** Plug each Z into the S_T equation to simulate thousands of different possible stock prices at expiration.
3. **Determine Payoff:** For each simulated stock price, calculate what the option contract would pay out (e.g., $\max(S_T - K, 0)$ for a Call option).
4. **Average and Discount:** According to the Law of Large Numbers, the simple average of all these simulated payoffs converges to the true mathematical expectation. We then discount this average back to the present day using the continuous risk-free rate (e^{-rT}) to find the fair price.

4.3 Python Implementation of Monte Carlo

Because Monte Carlo relies on independent statistical draws, it is perfectly suited for modern array-based computation. The following Python script utilizes NumPy vectorization to simulate 100,000 market paths and calculate the exact Black-Scholes price.

```

1 import numpy as np
2
3 # 1. Define Market and Option Parameters
4 S0 = 100.0      # Current stock price
5 K = 100.0      # Strike price
6 T = 1.0        # Time to maturity in years
7 r = 0.05       # Risk-free interest rate (5%)
8 sigma = 0.20   # Annualized volatility (20%)
9 num_simulations = 100000
10
11 # 2. Generate Random Market Shocks
12 # Draw 100,000 samples from a Standard Normal Distribution
13 Z = np.random.standard_normal(num_simulations)
14
15 # 3. Simulate Stock Prices at Expiration (T)
16 # This calculates all 100,000 paths simultaneously using vectorization
17 ST = S0 * np.exp((r - 0.5 * sigma**2) * T + sigma * np.sqrt(T) * Z)
18
19 # 4. Calculate Option Payoffs
20 # A Call option pays out the difference if S > K, otherwise 0
21 payoffs = np.maximum(ST - K, 0)
22
23 # 5. Calculate Expected Value and Discount to Present
24 # Average the payoffs (Law of Large Numbers) and discount by e^(-rT)
25 expected_payoff = np.mean(payoffs)
26 call_price = np.exp(-r * T) * expected_payoff
27
28 print(f"Simulated Call Option Price: ${call_price:.4f}")

```

Listing 1: Monte Carlo Simulation for Black-Scholes Pricing

4.4 Empirical Convergence Analysis

To empirically validate the computational robustness of the Monte Carlo approach, we execute the simulation framework across an exponentially increasing number of paths (N). As illustrated in Figure 1, the simulated option price exhibits high variance at lower trial counts.

However, in strict accordance with the Law of Large Numbers, as N approaches 10^5 , the standard error (represented by the 95% confidence intervals) shrinks proportionally to $1/\sqrt{N}$. Consequently, the Monte Carlo estimate aggressively converges upon the exact, closed-form analytical Black-Scholes price derived in Section 3, validating the integration of the risk-neutral random walk.

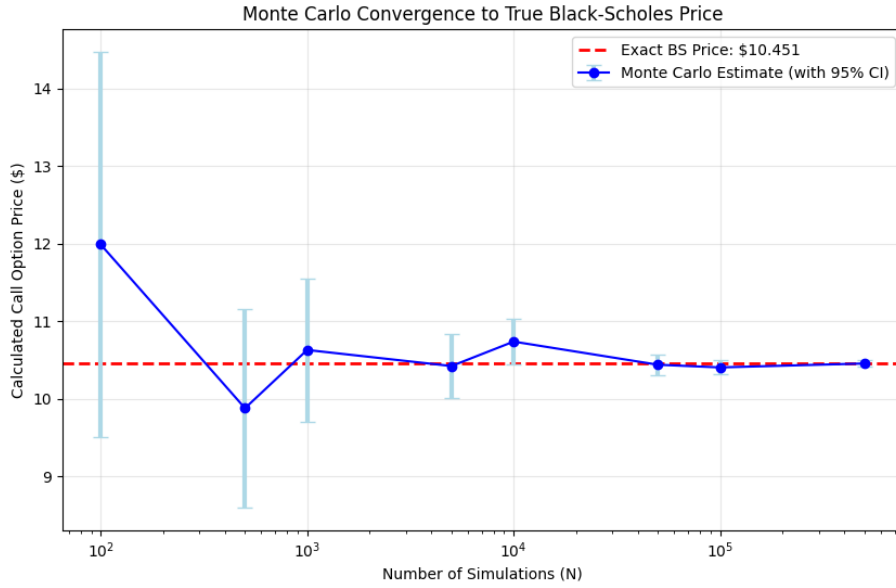


Figure 1: Convergence of the Monte Carlo simulation to the exact analytical Black-Scholes price (10.45) as the number of simulated paths (N) increases logarithmically. Error bars represent the 95% confidence interval.

5 Conclusion

This report has successfully traversed the rigorous mathematical architecture required to price financial derivatives in a continuous-time framework. We began by establishing the necessity of stochastic calculus, demonstrating how Geometric Brownian Motion and Ito's Lemma can effectively model the unpredictable, non-differentiable nature of asset prices while preserving the reality of limited liability.

By applying the economic principle of no-arbitrage to a continuously hedged, self-financing portfolio, we successfully eliminated market risk to derive the foundational Black-Scholes Partial Differential Equation. The primary mathematical achievement of this report is detailed in Section 3, where we demonstrated the exact closed-form analytical solution to this PDE. Through a series of domain transformations, we reduced the complex financial equation to the fundamental 1D Heat Equation. By utilizing the spatial Fourier Transform, we rigorously proved that a deterministic point source diffuses exactly as a log-normal probability density function over time. Furthermore, by invoking the sifting property of the Dirac delta distribution and leveraging the linearity of the differential operator, we mathematically justified the use of the convolution integral to evaluate the European Call option payoff.

Finally, we moved beyond theoretical proofs to construct a robust computational model. By directly integrating the risk-neutral random walk and applying the Law of Large Numbers, we implemented a vectorized Monte Carlo simulation in Python. The empirical analysis confirms that as the number of simulated paths increases, the computational model aggressively and accurately converges to the exact analytical price derived via the Heat Equation, thereby validating the integrity of the complete mathematical framework.

6 References

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